MASTER FUNCTIONAL SKILLS

MasterFunctionalSkills.co.uk | Official MFS Model Solutions

NCFE · LEVEL 1 · MATHEMATICS

NCFE Functional Skills Mathematics Level 1

Practice 1 Section B · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: NCFE-Maths-Level-1-Practice-1-Section-B

100% Built from Empty Question Paper + Official Mark Scheme

Verified by Tutor Abdul · MasterFunctionalSkills.co.uk

HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty NCFE Question Paper(s)
- Stepped MFS Multi-Colour Working & Process Marks
- Verified 100% Against Official NCFE Mark Scheme

© MasterFunctionalSkills.co.uk | Clean Empty-QP Model Solution Series

MASTER FUNCTIONAL SKILLS

ONLINE LEARNING PLATFORM & ACCREDITED COURSES

Pearson Edexcel & Open Awards Accredited Suite · Entry 3, Level 1 & Level 2

Pass Your Functional Skills Exams With Confidence

100% Pass Guarantee | Dyslexia-Friendly Learning | 1-to-1 Expert Tutor Support

Master Functional Skills provides an adaptive, dyslexia-friendly online learning platform tailored for adult learners, apprentices, and healthcare & vocational staff. Everything you need to revise, practice, and pass is at your fingertips.

1. Smart Diagnostic Initial Assessment

Calculate your exact working level and pinpoint knowledge gaps instantly

- **Exact Grade Calculation:** Adaptive assessments calculate your exact level across all topics.
- **Personalised Study Pathway:** Skip what you know; focus 100% on the topics needed to pass.
- **Real-Time Exam Readiness:** Predictive tracker confirms when you are 100% ready to book.

2. Comprehensive Video Lessons & Worked Solutions

Bite-sized visual explainers covering every single Maths & English curriculum standard

- **Step-by-Step Video Lessons:** Master percentages, algebra, perimeter, ratio, and reverse fractions.
- **Multi-Ink Handwriting Models:** Visual 3-colour methods showing working, carries, and examiner tips.
- **Tutor Shortcuts & Golden Rules:** Proven non-calculator mental arithmetic hacks and BIDMAS tricks.

3. Full Mock Exams & Real-Time Performance Analytics

Practice in exam conditions and watch your scores and confidence climb

- **Authentic Exam Papers:** Timed Section A & Section B papers formatted exactly like Pearson Edexcel.
- **Instant Topic Breakdown:** View percentage scores across Number, Measure/Shape, and Data.
- **Historical Score Tracking:** Monitor past attempts to ensure continuous improvement before exam day.

Ready to Master Your Functional Skills & Pass First Time?

Join thousands of healthcare staff, apprentices, and adult learners across the UK.

Instant 24/7 access on desktop, tablet, and mobile with no software to install.

START LEARNING TODAY AT

MasterFunctionalSkills.co.uk

Examiner use only

Activity 2: Caring for reptiles

2 (a) Zac is doing an apprenticeship in Animal Care and Management.

He is working as an assistant at a reptile centre.

Zac learns about how to look after reptiles.

Habitat	Animal type	Maximum length in centimetres (cm)	Optimal temperature	Humidity
Water	Box terrapin	15 cm	20 °C to 25 °C	80% to 100%
Land	Garter snake	75 cm	22 °C to 27 °C	60% to 80%

Q2(a) Worked Solution:
150 (mm)
[1 mark]

What is the maximum length of a Box terrapin in millimetres (mm)?

[1 mark]

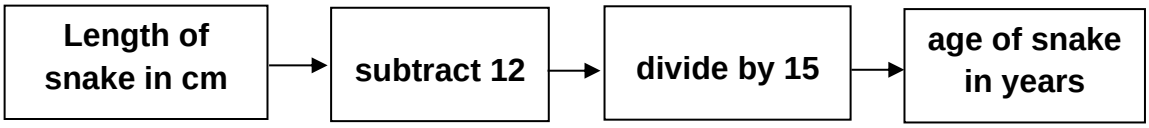
Your answer: **mm**

past paper

Please turn over

Examiner use only

2 (b) The age of a young Garter snake can be estimated using this rule:



An adult Garter snake is about 72 cm long.

Q2(b) Worked Solution:

What age is a Garter snake when its length is 72 cm?

[3 marks]

4 (years)
72 – 12 or 60
Their 60 ÷ 15
[3 marks]
Tip: Award 3 marks if correct answer given

Your answer: **years**

past paper

Examiner use only

2 (c) It is important that reptiles are given the correct diet.

The calcium : phosphorus weight ratio in their food should be close to 2 : 1

The table shows the weight in milligrams (mg) of calcium and phosphorus in one scoop of food.

Q2(c) Worked Solution:
4 : 1 and 2 : 1 seen
Cauliflower with 4 : 1 and 2 : 1 seen
[2 marks]

		Calcium	Phosphorus
Food	**Watermelon**	36 mg	9 mg
	Cauliflower	160 mg	80 mg

Which food has the better calcium : phosphorus ratio for reptiles?
Show your working.

[2 marks]

Your answer:

past paper

Please turn over

Examiner use only

2 (d) A customer needs to buy a tank for his snake.

Zac has this information about the sizes of tanks:

Length of snake (to nearest cm)	Minimum floor area for one snake (cm^2)
0 to 30	300
31 to 40	400
41 to 50	600
51 to 75	1200
over 75	2500

The reptile centre sells these five sizes of tanks.

Q2(d) Worked Solution:

Size of tank	Length (cm)	Width (cm)	Price
Mini	18	11	£3.15
Small	23	15.5	£6.22
Medium	31	19.5	£9.95
Large	37	22	£13.99
Extra large	45	30	£18.99

Calculates any area correctly

Which is the cheapest tank the customer should buy for one snake which is 45 cm long? Show your working.

[3 marks]

23 × 15.5 or 356.5
and 31 × 19.5 or 604.5
Medium or 31 × 19.5 or £9.95 stated
[3 marks]
Tip: Accept use of 70

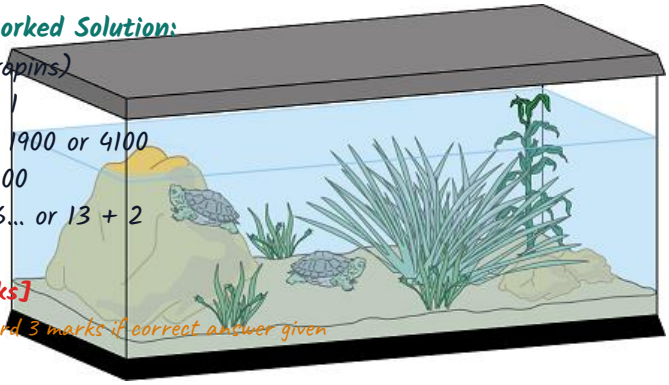
Your answer:

past paper

Examiner use only

2 (e) Another customer already has a terrapin tank.
She wants advice on how many terrapins to buy.

Q2(e) Worked Solution:
15 (terrapins)
method 1
6000 – 1900 or 4100
their 4100
or 13.66... or 13 + 2
300
[3 marks]
Tip: Award 3 marks if correct answer given



When her tank is filled, the water surface area is 6000 cm^2

Terrapins need a water surface area of:

- 1900 cm^2 for two terrapins
- then an additional 300 cm^2 for each additional terrapin.

What is the maximum number of terrapins Zac should advise the customer to buy?

[3 marks]

Your answer: **terrapins**

past paper

Please turn over

Examiner use only

2 (f) At college, Zac learns about snakes.

He reads this data about the lengths of snakes found in one area:

Length of snake (to nearest cm)	Number of snakes
20–24	7
25–29	11
30–34	13
35–39	8
40–44	6
45–49	4

Q2(f) Worked Solution:

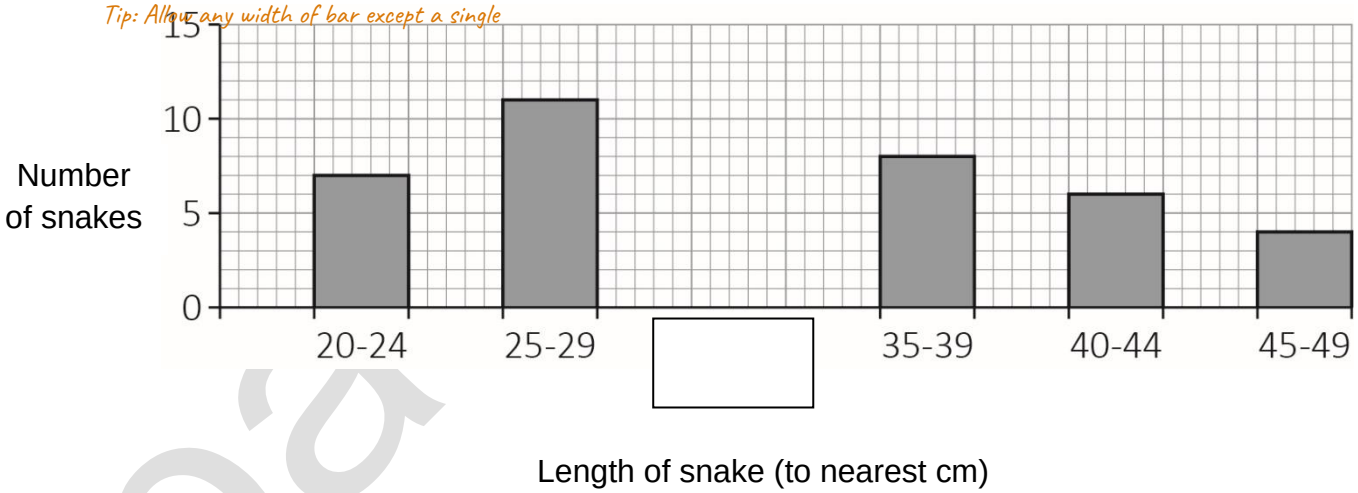
The bar chart shows this data but one bar is incomplete.

Draw the bar to the correct height on the chart and label it.

[2 marks]

30-34 correctly labelled
Bar drawn to 13
4
[2 marks]

Tip: Allow any width of bar except a single



paper

Examiner use only

2 (g) The class tutor has provided some snakes for the students to handle.

5 of the snakes are green and 4 are brown.

Zac is first to choose and picks a snake at random.

What is the probability that he picks a brown snake?

[1 mark]

Q2(g) Worked Solution:
accept 0.44 or 44%
9
trip (Calculator Test)
[1 mark]

Your answer:

[Total marks: 15]

past paper

Please turn over

Examiner use only

Activity 3: Cycling trip

3 (a) Asha lives in Nottingham. She is planning a cycling trip in Holland.

Asha plans the trip in two stages:

- take the train from Nottingham to Harwich
- take the ferry from Harwich to the Hook of Holland.

This is part of the train timetable from Nottingham to Harwich:

Q3(a) Worked Solution:
/
[Method & Accuracy]

Depart	From	To	Arrive
17:02	Nottingham (NOT)	Harwich International (HPQ)	21:17
18:37	Nottingham (NOT)	Harwich International (HPQ)	22:17
19:28	Nottingham (NOT)	Harwich International (HPQ)	22:58
20:34	Nottingham (NOT)	Harwich International (HPQ)	23:53

How many hours does the 19:28 train from Nottingham take to get to Harwich if it runs to time? Give your answer as a mixed number.

[2 marks]

Your answer: **hours**

past paper

Examiner use only

3 (b) The return train fare from Nottingham to Harwich is £158.80

Asha has a student railcard which gives her 30% off.

How much will Asha pay for her return ticket?

[2 marks]

Q3(b) Worked Solution:

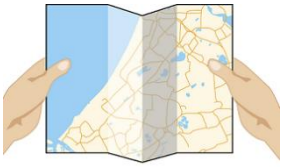
Full method & answer verified against official NCFE mark scheme.

[Method & Accuracy]

Your answer: **£**

3 (c) Asha plans to cycle from the Hook of Holland to Amsterdam.

Her map has a scale of 1 cm to 8 (kilometres) km



The straight line distance between the Hook of Holland and Amsterdam is 70 km

Q3(c) Worked Solution:

What is the measurement between these two places on Asha's map?
Give your answer in mm

[2 marks]

Full method & answer verified against official NCFE mark scheme.

[Method & Accuracy]

Your answer: **mm**

past paper

Please turn over

Examiner use only

3 (d) Asha will spend two days in Amsterdam and then cycle back to catch the ferry at the Hook of Holland.

The road distance between two places is about 40% longer than the straight line distance.

Estimate the actual road distance Asha will cycle **in total**.
Give your answer in km

[3 marks]

Q3(d) Worked Solution:

Full method & answer verified against official NCFE mark scheme.

[Method & Accuracy]

Your answer: **km**

past paper

Examiner use only

3 (e) Asha needs to buy a pair of cycle bags for her trip.

She reads an article which says that, for a cycling holiday, the cycle bags need to have a capacity of at least 20 litres each.

1 litre is the same as 1000 cm^3

Asha sees some cycle bags which each have these dimensions:

- height 42 cm
- width 32 cm
- depth 17 cm



The cycle bags are roughly cuboid in shape.

Are these cycle bags large enough?
Show your working.

[3 marks]

Q3(e) Worked Solution:
Yes and ~~22.8(48...) or 23~~
[Method & Accuracy]

Your answer:

past pap

Please turn over

Examiner use only

3 (f) The empty cycle bags weigh 0.76 kg each.

Asha says, “Two of these cycle bags together will weigh less than 1.6 kg.”

Is Asha correct? Explain your answer.

[1 mark]

Q3(f) Worked Solution:

Full method & answer verified against official NCFE mark scheme.

[Method & Accuracy]

Your answer:

3 (g) The cycle bags that Asha buys weigh 0.85 kg each.

Asha fills one bag with clothes.

The clothes weigh 473 grams (g).

How much does the cycle bag containing the clothes weigh?

[1 mark]

Q3(g) Worked Solution:

Full method & answer verified against official NCFE mark scheme.

[Method & Accuracy]

Your answer:

past paper

Examiner use only

3 (h) Asha's instructions say, "When you leave the ferry port, go east."

Asha is cycling north.

How many degrees clockwise does she need to turn to cycle east?

[1 mark]

Q3(h) ~~Worked Solution:~~

Full method & answer verified against official NCFE mark scheme.

[Method & Accuracy]

Your answer: **degrees**

[Total marks: 15]

past paper

Please turn over

Examiner use only

Activity 4: Sleeping rough

4 (a) Every year, local authorities in England estimate how many people are sleeping rough in their area.

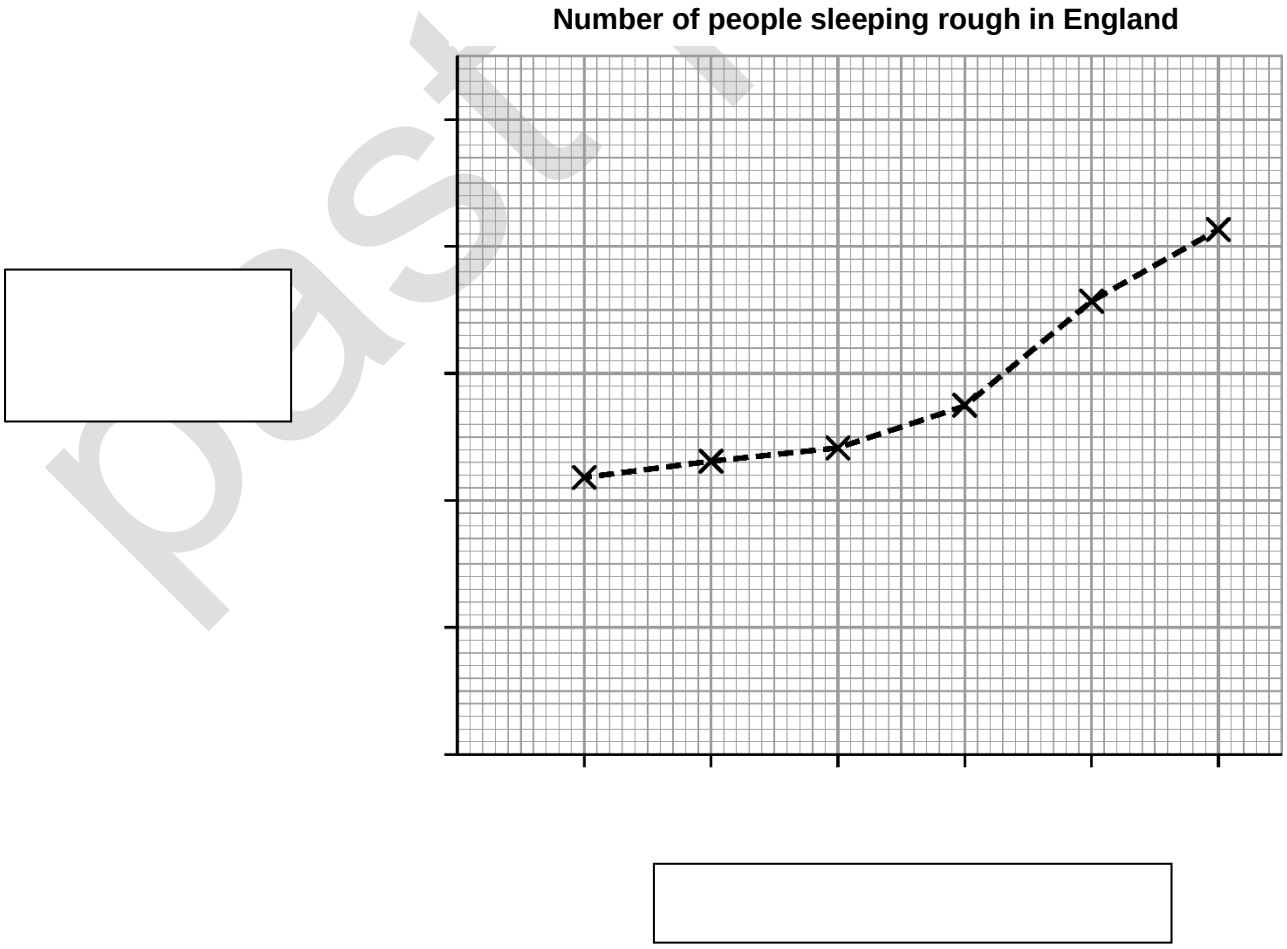
The estimates for England over the last few years are shown in this table:

Year	Estimated number of people sleeping rough in England
2011	2181
2012	2309
2013	2414
2014	2744
2015	3569
2016	4134

The data is represented in this line graph. The axes are incomplete.

Complete the line graph.

[3 marks]



past paper

Question 4(a) · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

Q4(a) Worked Solution:

Vertical scale correctly completed
0, 1000, 2000, 3000, 4000, 5000
Horizontal scale correctly completed
2011, 2012, 2013, 2014, 2015, 2016
Both axes correctly labelled
[3 marks]

Examiner use only

4 (b) Between 2014 and 2017 the number of people sleeping rough increased by 75%

Write this percentage as a fraction in its simplest form.

[1 mark]

Q4(b) Worked Solution:
3
4
[1 mark]

Your answer:

4 (c) Calculate 75% of 2744
Give your answer to the nearest hundred.

[2 marks]

Q4(c) Worked Solution:
2100 (people)
75
3
x2744
□ 2744 or or 2058
100
4
[2 marks]
Tip: Award 2 marks if correct answer given

Your answer:

past paper

Please turn over

Examiner use only

4 (d) A charity asks people to donate 40p a day to sponsor rooms for people sleeping rough.

The charity hopes to raise 1 million pounds a year through these donations.

How many people would have to donate 40p **a day** to raise at least £1 000 000 for the charity in **a year**?

Use the conversion: 365 days = 1 year

[3 marks]

Q4(d) Worked Solution:

6850 (people)

1

7

40

or 146

□ 365

100

1000000 ÷ their 146

[3 marks]

Tip: Award 3 marks if correct answer given

Your answer:

past paper

Examiner use only

4 (e) A community group raises money to help people who sleep rough.

Volunteers pack shopping bags at a local supermarket and ask shoppers for donations.

The table shows the numbers of volunteers who packed bags on 32 Saturdays in 1 year.

Q4(e) Worked Solution:

4	**5**	**2**	**4**	**6**	**5**	**5**	**4**
2	**2**	**1**	**5**	**5**	**3**	**4**	**3**
3	**4**	**2**	**2**	**5**	**2**	**1**	**0**
3	**4**	**4**	**4**	**3**	**1**	**1**	**5**

6, 5, 8 correctly placed

Use the data to complete this frequency table.

[2 marks]

2 or 3 correct frequencies

3 correct frequencies

[2 marks]

Number of volunteers	Frequency
0	1
1	4
2	
3	
4	
5	7
6	1
Total	**32**

pastpaper

Please turn over

Examiner use only

4 (f) Over these days the volunteers collected £6224

What was the mean amount of money collected each Saturday?

[2 marks]

Q4(f) Worked Solution:
(£)194.50
6224
or 194.5 seen
32
[2 marks]
Tip: Award 2 marks if correct answer given

Your answer: **£**

4 (g) The amounts of money collected each Saturday had a range of £53

The most money collected in a day was £220

What was the least amount of money collected in a day?

[1 mark]

Q4(g) Worked Solution:
(£)167
[1 mark]

Your answer: **£**

past paper

Examiner use only

4 (h) Abdhul is a member of the community group.

On a Saturday he either packs shopping bags or he takes part in another activity.

The table shows the probability of him taking part in each activity.

Q4(h) Worked Solution:

Activity	Probability
Bag-packing	0.35
Football	0.50
Computer games	0.11
Shopping	0.04

Football because it or 0.50 is the

Which activity is Abdul most likely to take part in?
Explain how you decide.

[1 mark]

biggest (probability)
8
[1 mark]
Tip: Accept highest, biggest, largest (not most

Your answer:

[Total marks: 15]

This is the end of the assessment.

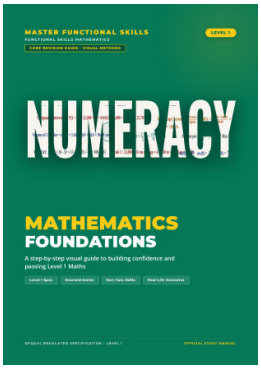
past paper

MASTER FUNCTIONAL SKILLS STORE

OFFICIAL REVISION GUIDES, WORKBOOKS & FLASHCARDS

Written by accredited specialist tutors to match the exact 2026 Pearson Edexcel standards

Everything You Need To Revise, Practice & Pass Your Maths Level 1 Exam

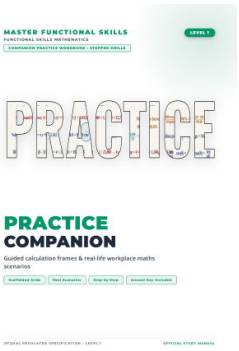


Core Curriculum Manual

Maths Level 1
Core Study Guide

- Full official curriculum coverage
- Section A Non-Calc & Sec B Calc
- Worked models & golden rules
- Dyslexia-friendly visual format

Spiral Bound Print & PDF

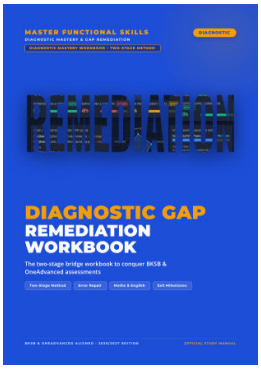


1,000+ Practice Drills

Maths Level 1
Companion Workbook

- 1,000+ exam-style questions
- Progressive difficulty drills
- Full worked step-by-step key
- Targeted fix for weak areas

Spiral Bound Print & PDF



Realistic Mock Papers

Maths Level 1
Official Mock Packs

- Full mock exam paper sets
- Authentic Pearson Edexcel style
- Detailed mark schemes included
- Grade boundaries & pass marks

Exam Booklet Pack & PDF



Pocket Flashcards A6

Maths Level 1
Pocket Revision Cards

- Compact A6 cards for quick study
- Essential formulas & conversions
- Dual-sided question & answer
- High-yield mental arithmetic rules

Boxed Card Deck & Print PDF

Order Official Study Guides & Revision Cards Online

Free Next-Day UK Tracked Shipping · Instant PDF Downloads Included

Trusted by training providers, apprentices & adult learners nationwide.

VISIT THE ONLINE STORE

MasterFunctionalSkills.co.uk